

Unit-2: Modulation and Switching Techniques

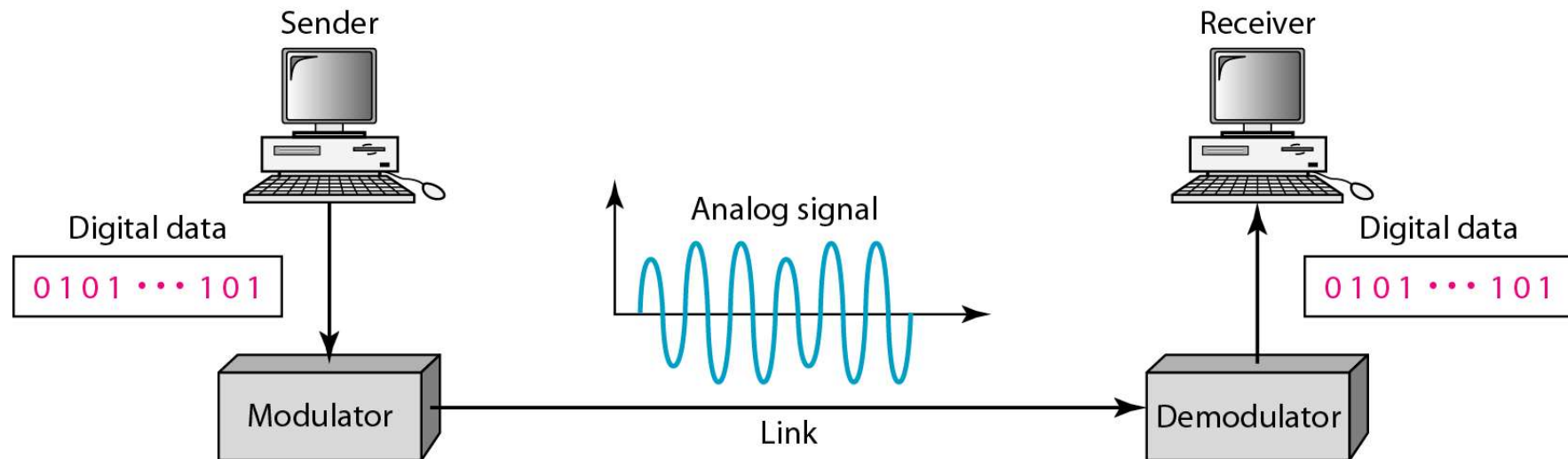
Unit-2: Modulation and Switching

Analog Transmission: Modulation of Digital Data, Modulation of Analog signals. Multiplexing: FDM, WDM, TDM, Transmission Media: Guided Media, Unguided Media, Circuit Switching and Packet Switching.

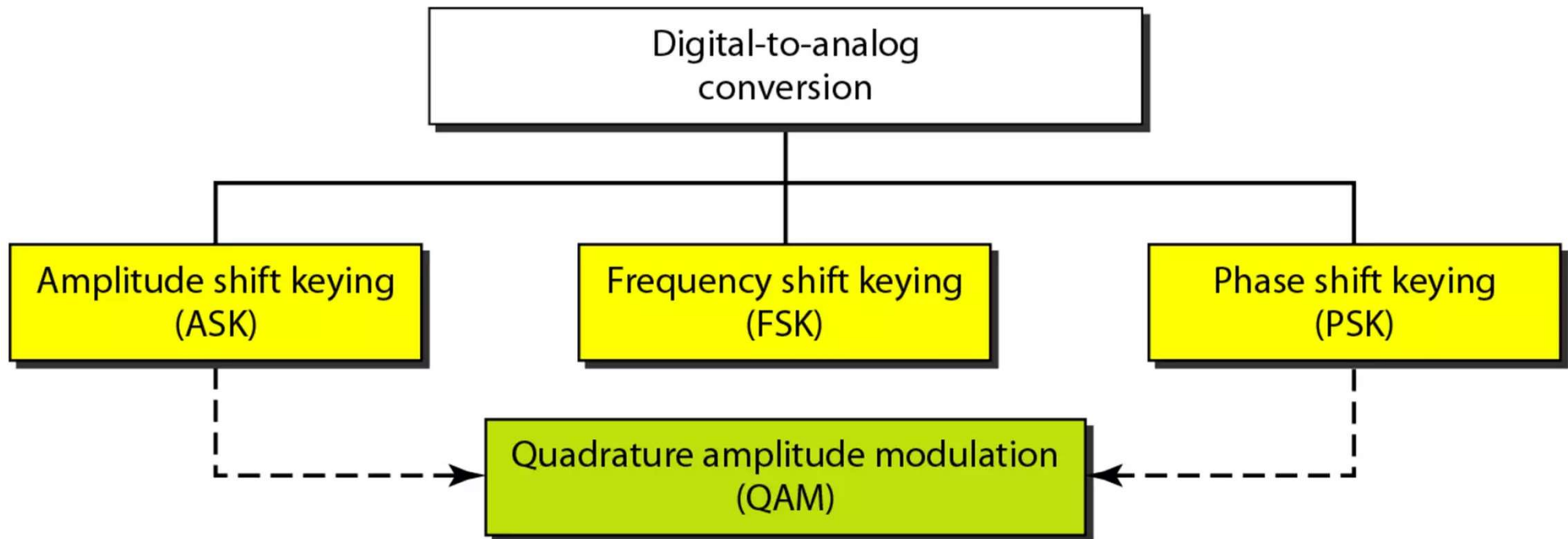
Analog Transmission

Digital to Analog Conversion

- Digital-to-analog conversion is the process of changing one of the characteristics of an analog signal based on the information in digital data.
- Digital data needs to be carried on an analog signal.
- A carrier signal (frequency f_c) performs the function of transporting the digital data in an analog waveform.



Types of digital-to-analog conversion



Note

- ✓ Bit rate, N , is the number of bits per second (bps).
- ✓ Baud rate is the number of signal elements per second (bauds).
- ✓ In the analog transmission of digital data, the signal or baud rate is less than or equal to the bit rate.

$$S = N \times 1/r \text{ bauds}$$

r is the number of data bits per signal element.

Q1. An analog signal carries 4 bits per signal element. If 1000 signal elements are sent per second. Find the bit rate.

Solution: In this case, $r=4$, $S=1000$

$$S = N \times 1/r$$

$$\text{Hence, } N = S \times r = 1000 \times 4 = 4000\text{bps}$$

Amplitude Shift Keying (ASK)

- The general expression of an electrical signal is given by

$$C(t) = A \cos(2\pi f_c t + \phi)$$

A= Amplitude

f_c = Frequency

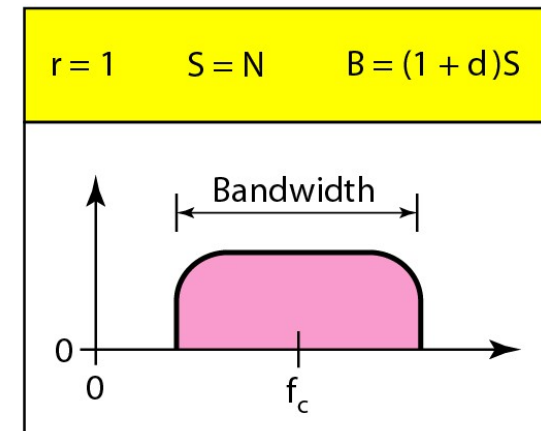
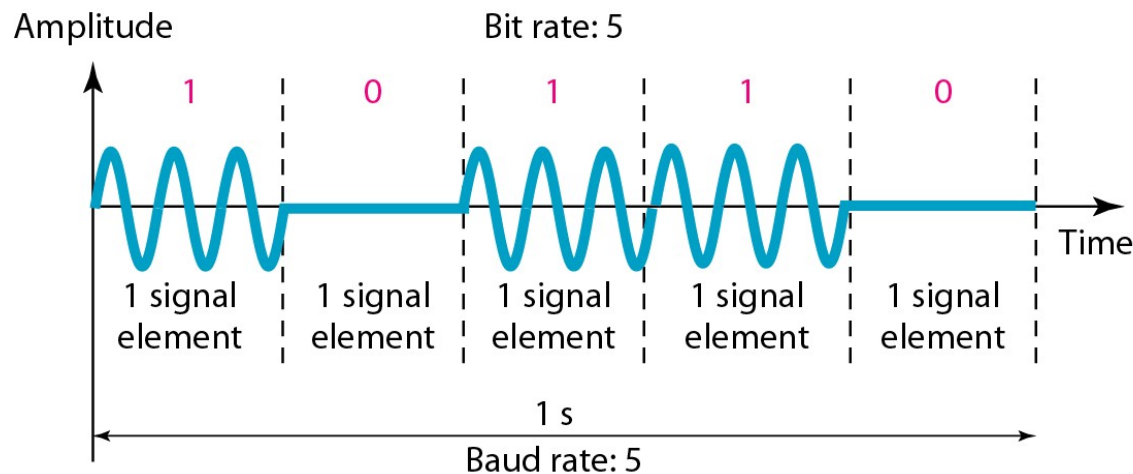
Φ = Phase angle

- In *ASK*, the amplitude of the carrier signal is varied to create signal elements.

Where as both frequency and phase remain constant.

Binary ASK

- ASK is implemented by changing the amplitude of a carrier signal to reflect amplitude levels in the digital signal.
- a digital “1” could not affect the signal, whereas a digital “0” would, by making it zero.

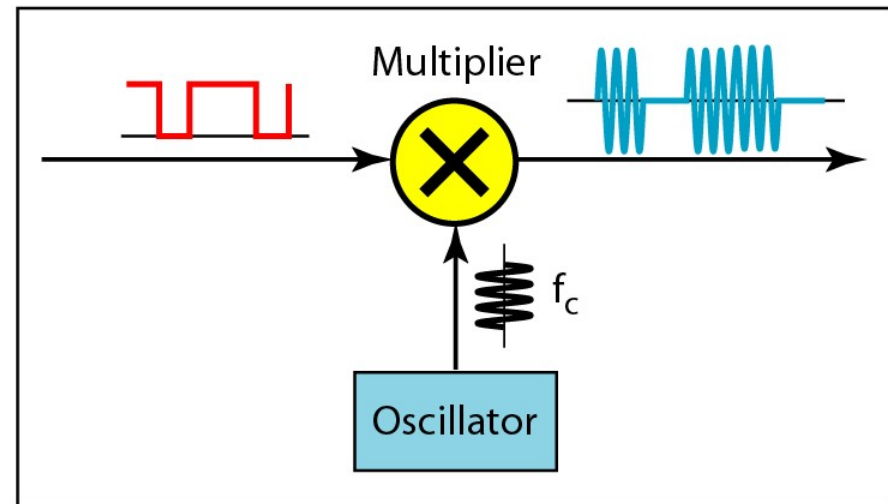
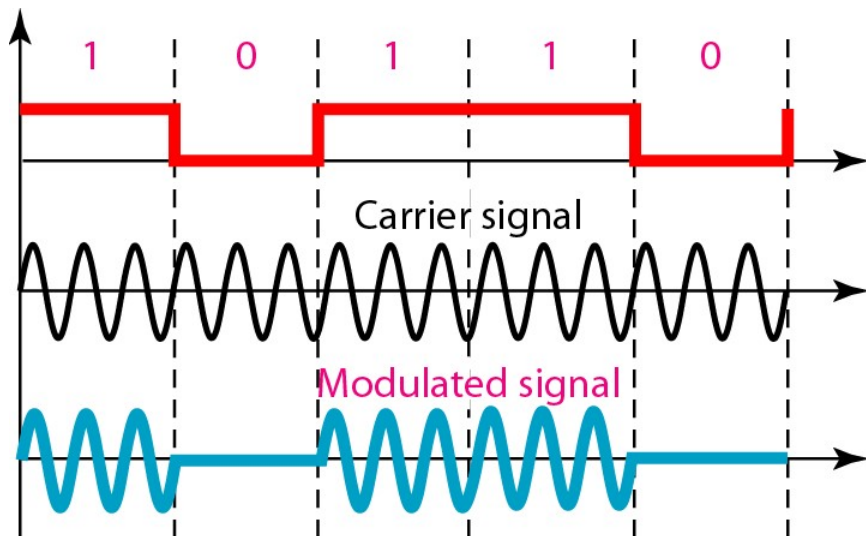


- The value of d is between 0 and 1. f_c is the carrier frequency which is located the middle of bandwidth
- Bandwidth: **$B = (1+d)S$**

Binary ASK

The line encoding will determine the values of the analog waveform to reflect the digital data being carried.

BASK modulated signal is generated by multiplying unipolar NRZ with carrier signal.



We have an available bandwidth of 100 kHz which spans from 200 to 300 kHz. What are the carrier frequency and the bit rate if we modulated our data by using ASK with $d = 1$?

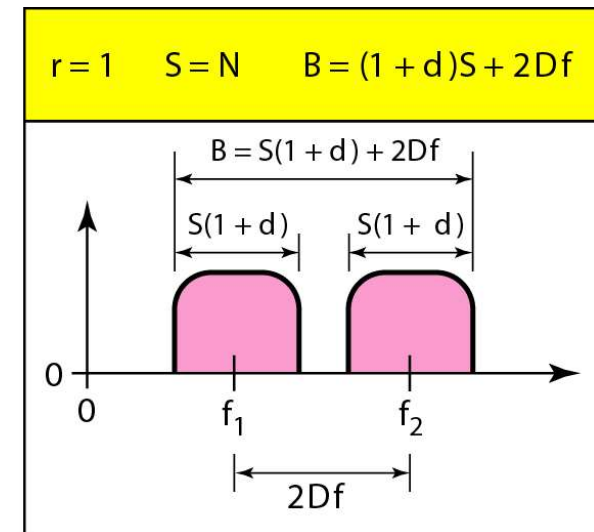
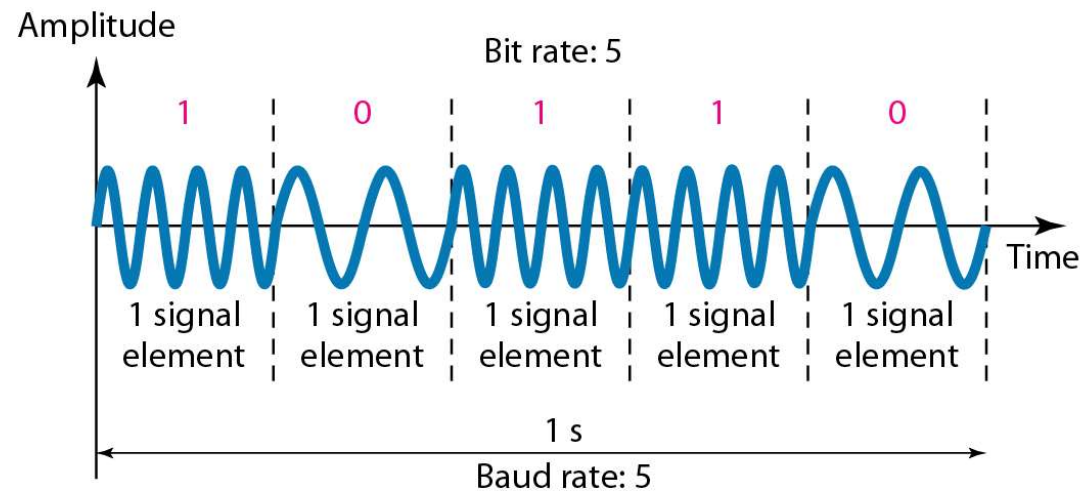
Solution

The middle of the bandwidth is located at 250 kHz. This means that our carrier frequency can be at $f_c = 250$ kHz. We can use the formula for bandwidth to find the bit rate (with $d = 1$ and $r = 1$).

$$B = (1 + d) \times S = 2 \times N \times \frac{1}{r} = 2 \times N = 100 \text{ kHz} \quad \rightarrow \quad N = 50 \text{ kbps}$$

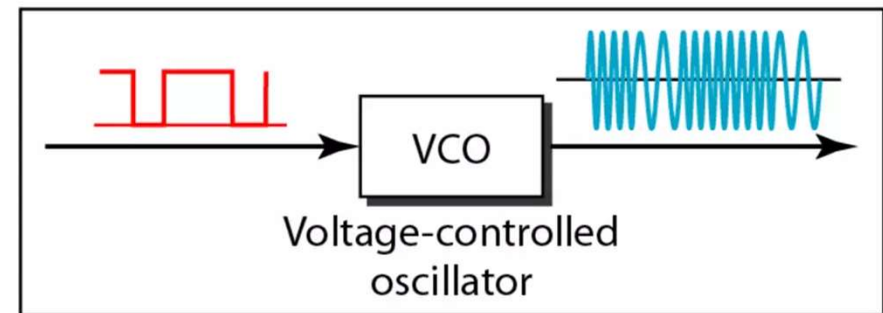
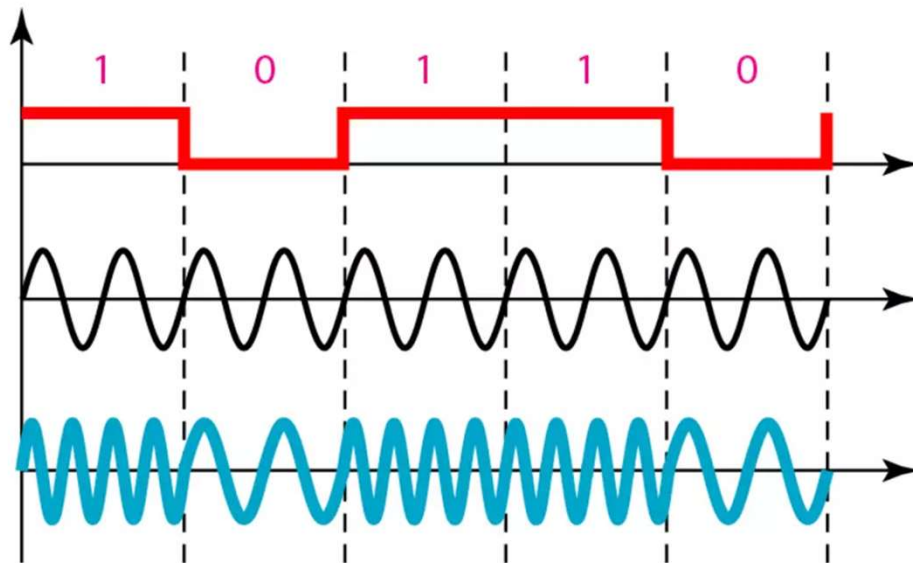
Frequency Shift Keying (FSK)

- In *FSK*, the frequency of the carrier signal is varied to represent data.
- For example, a “1” could be represented by $f_1 = f_c + \Delta f$, and a “0” could be represented by $f_2 = f_c - \Delta f$.



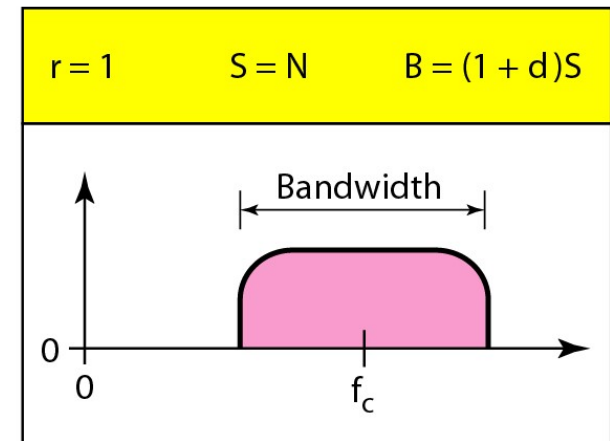
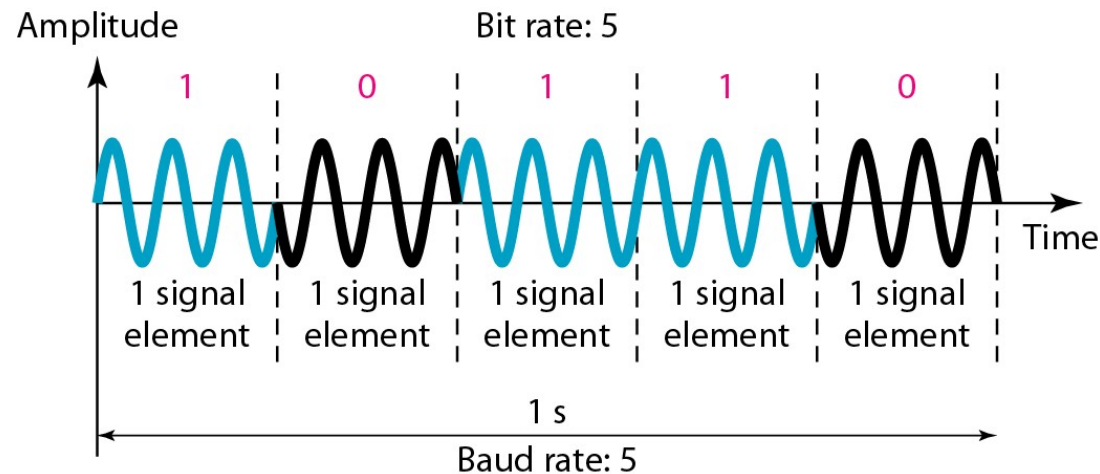
Implementation of BFSK

- BFSK can be implemented using NRZ and VCO
- A **VCO** is an oscillator whose **output frequency** depends on an **input control voltage**.



Phase Shift Keying (PSK)

- In PSK, the phase of the carrier is varied to represent two or more different signal elements.
- Noise can change the amplitude easier than it can change phase.
- PSK is superior to FSK because we do not need two carrier signals.
- PSK needs more sophisticated hardware compared to FSK to be able to distinguish between phases.



Binary Phase Shift Keying (BPSK)

BPSK modulated signal is generated by multiplying Polar NRZ with carrier signal.

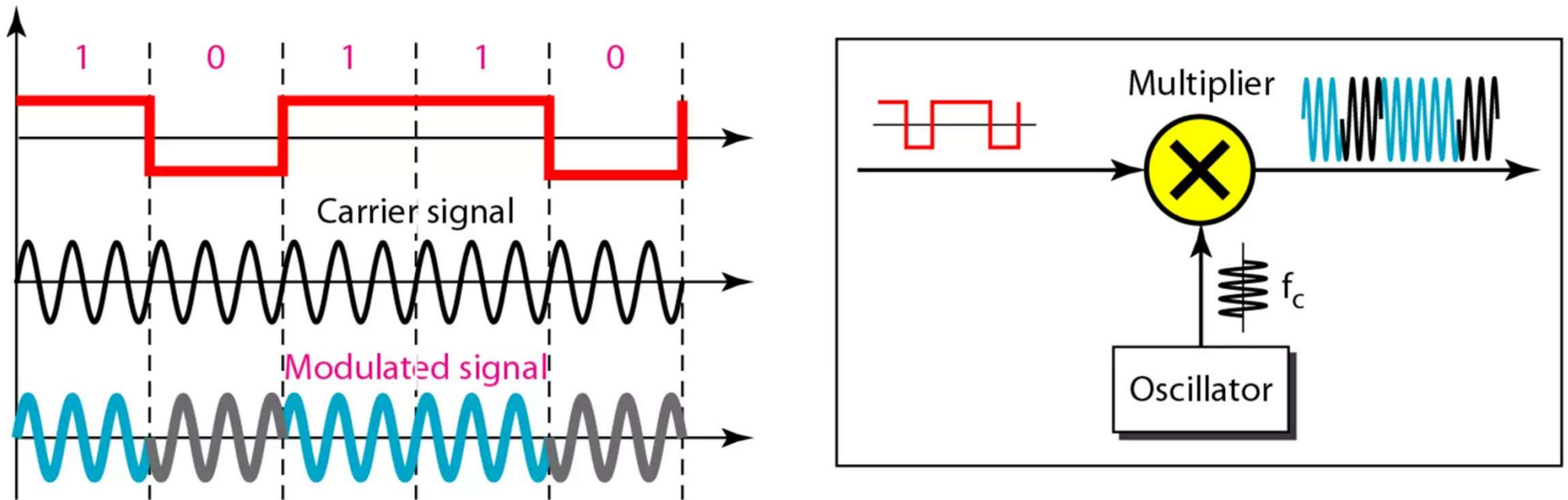
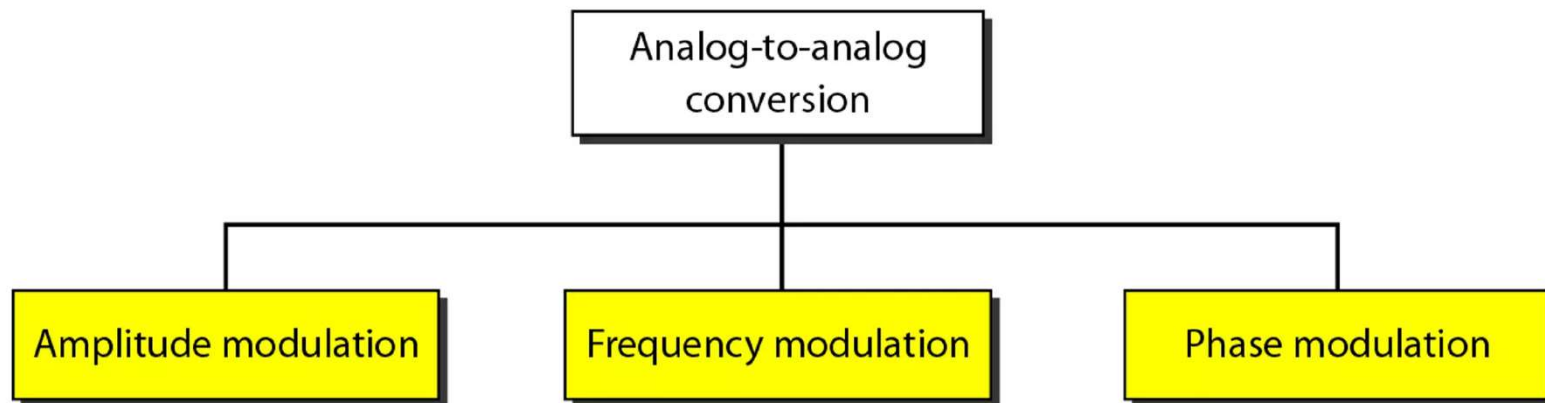


Figure: *Implementation of BPSK*

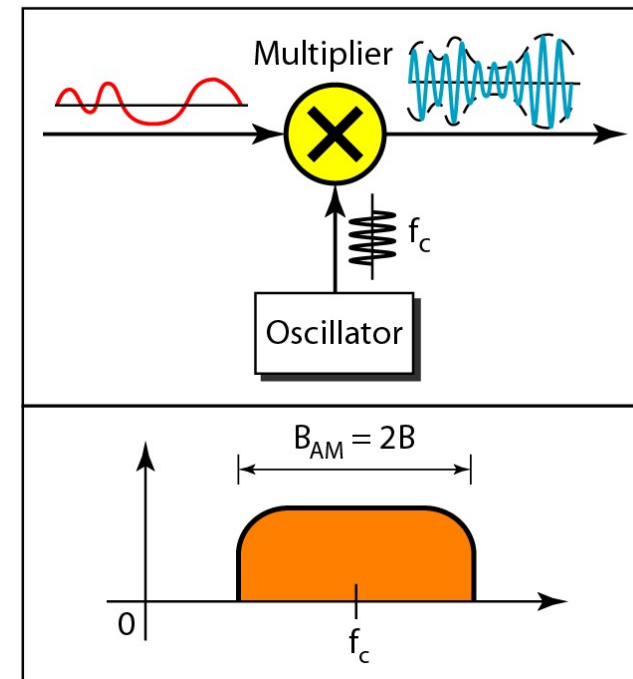
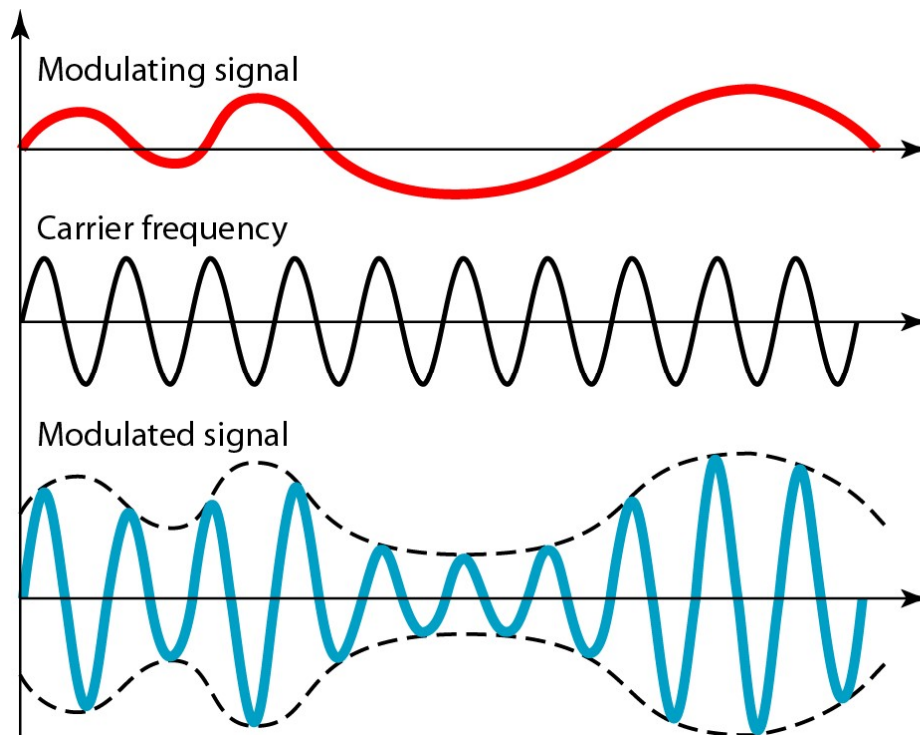
Analog to Analog Conversion

- Analog-to-analog conversion or analog modulation is the representation of analog information by an analog signal.
- One may ask why we need to modulate an analog signal; it is already analog.
- Modulation is needed if the medium is bandpass in nature or if only a bandpass channel is available to us.



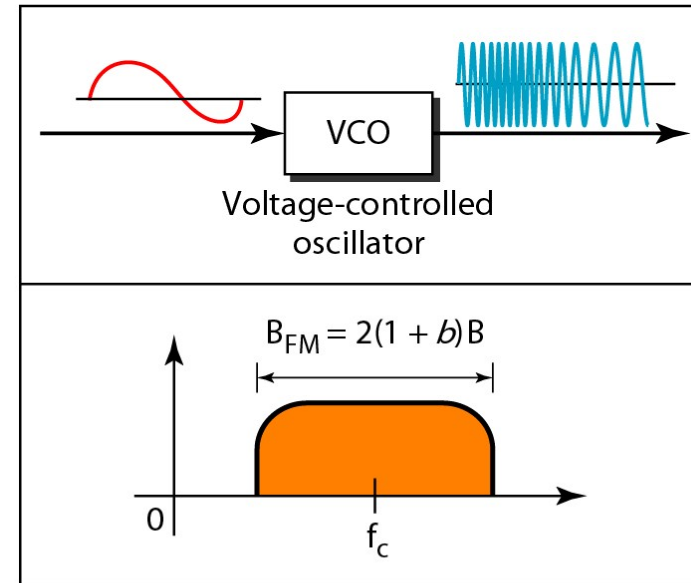
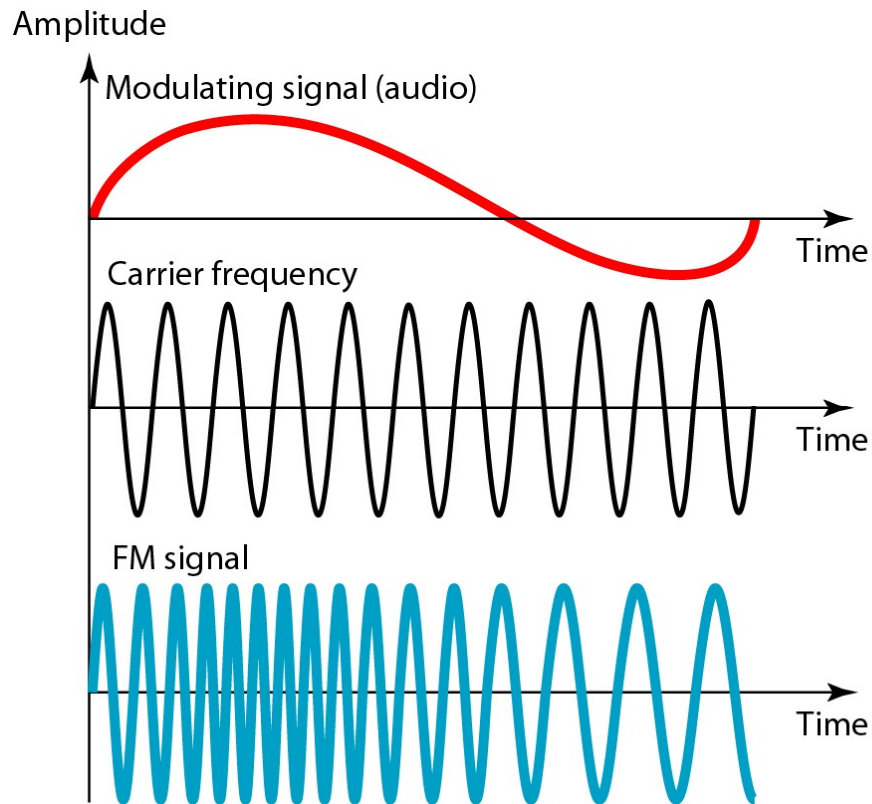
Amplitude Modulation (AM)

- In *AM transmission*, the *amplitude* of the carrier signal varies with the changing amplitudes of the modulating signal.
- The modulating signal is the envelope of the carrier.
- The required bandwidth is $2B$, where B is the bandwidth of the modulating signal.



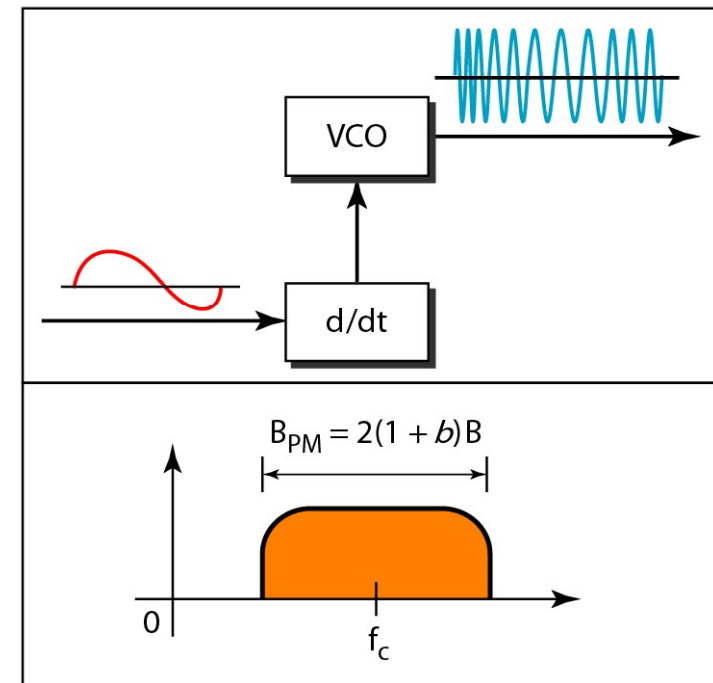
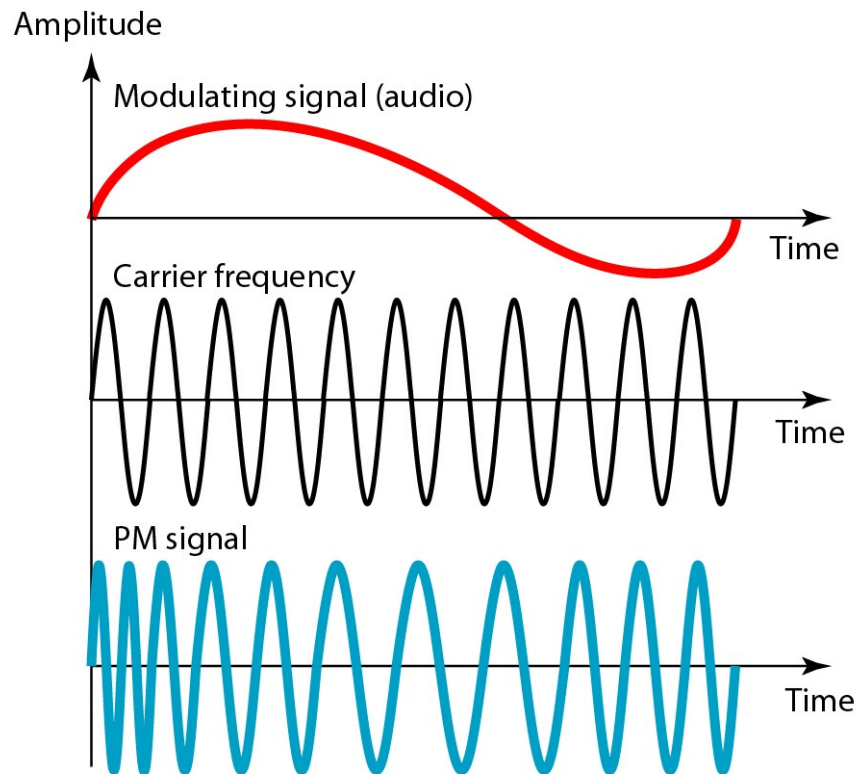
Frequency Modulation (FM)

- The modulating signal changes the *frequency* (f_c) of the carrier signal.
- The frequency of the oscillator changes according to the input voltage.



Phase Modulation (PM)

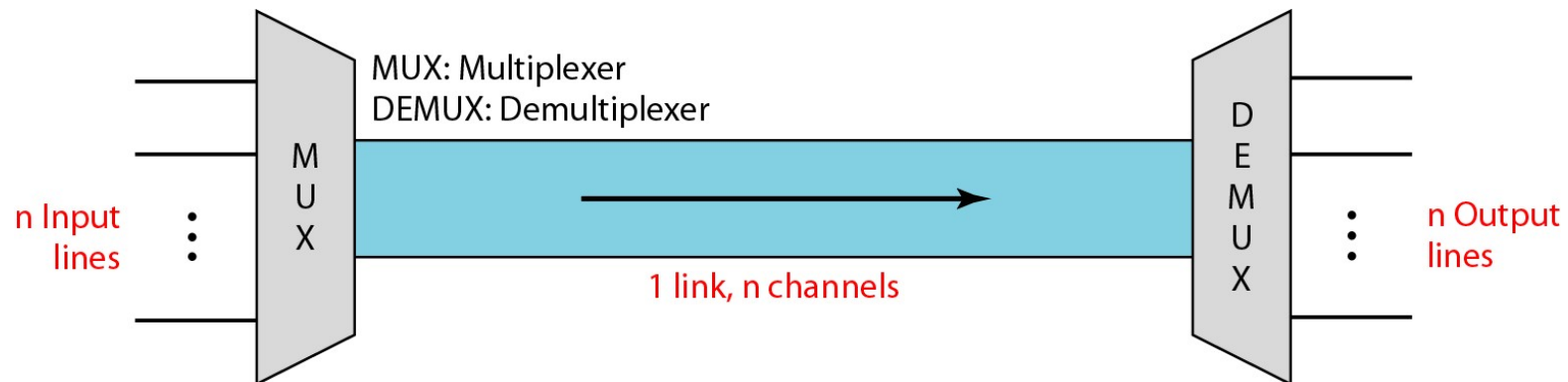
- The modulating signal only changes the phase of the carrier signal.
- The *phase change* manifests itself as a frequency change but the instantaneous frequency change is proportional to the derivative of the amplitude.



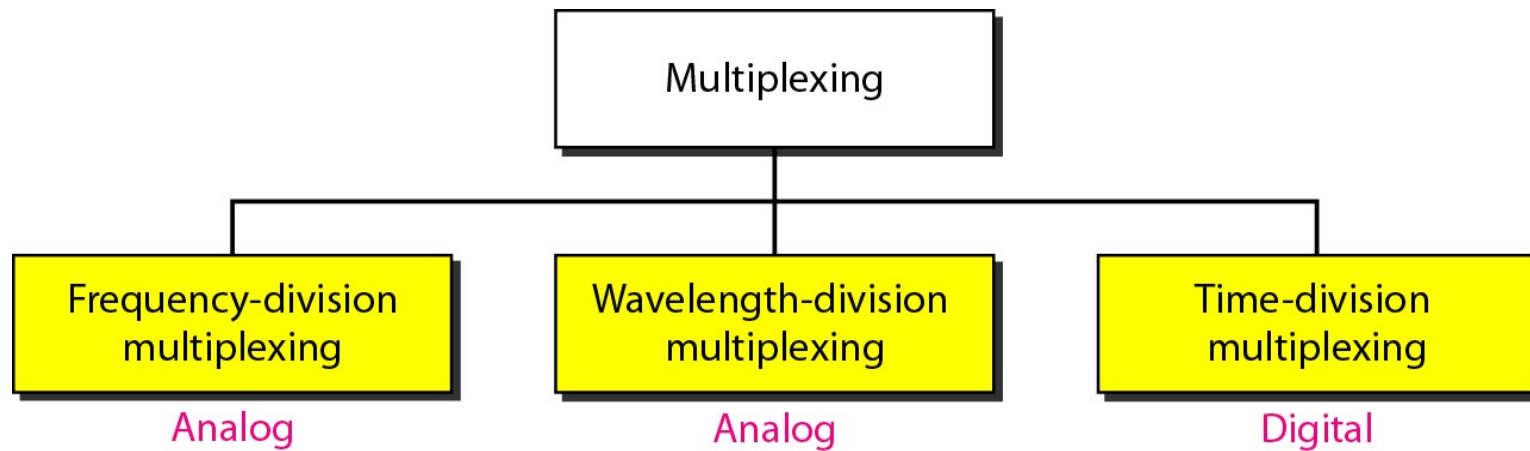
MULTIPLEXING

MULTIPLEXING

- **Multiplexing** is a technique that allow the simultaneous transmission of multiple signals across a single data link.
- **Multiplexer (MUX)**, combines n streams of data into a single stream (many-to-one).
- **Demultiplexer (DEMUX)**, which separates the stream back into its component transmissions (one-to-many) and directs them to their corresponding lines.



TYPES OF MULTIPLEXING

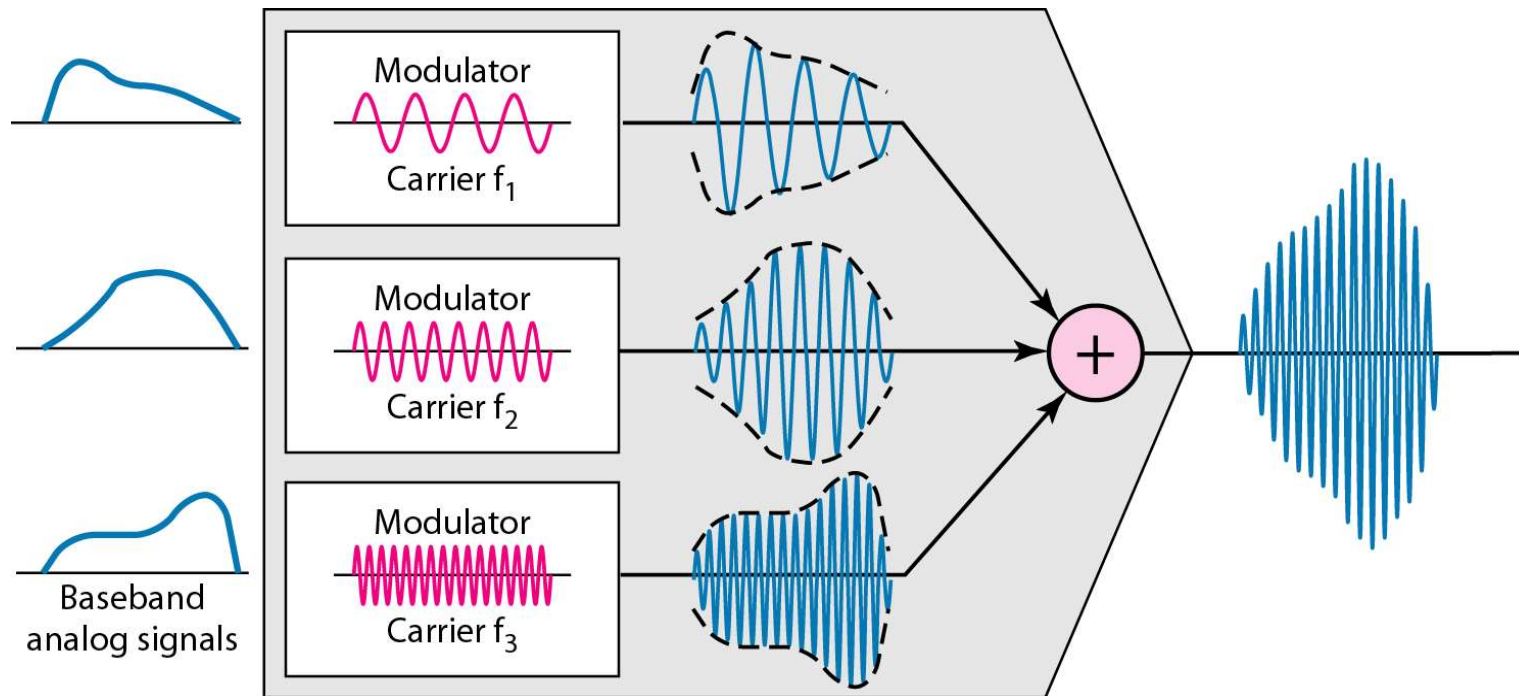


FREQUENCY DIVISION MULTIPLEXING (FDM)

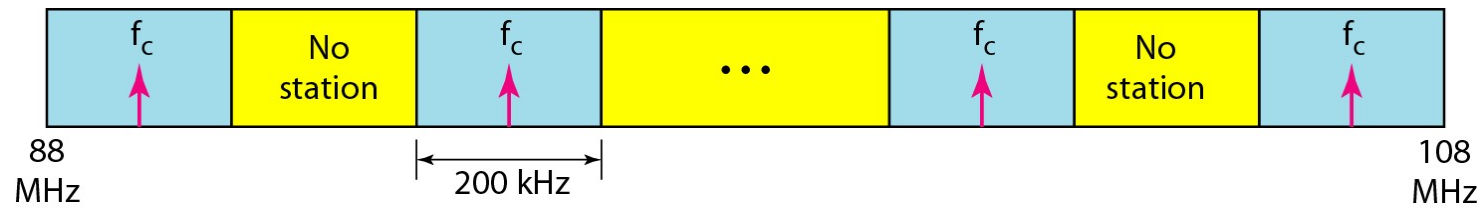
- FDM is an analog multiplexing technique that combines analog signals.
- A digital signal can be converted to an analog signal before FDM is used to multiplex them.
- **Guard bands:** to separate signals of different carrier frequencies.
- Use: Telephone lines, FM, AM.



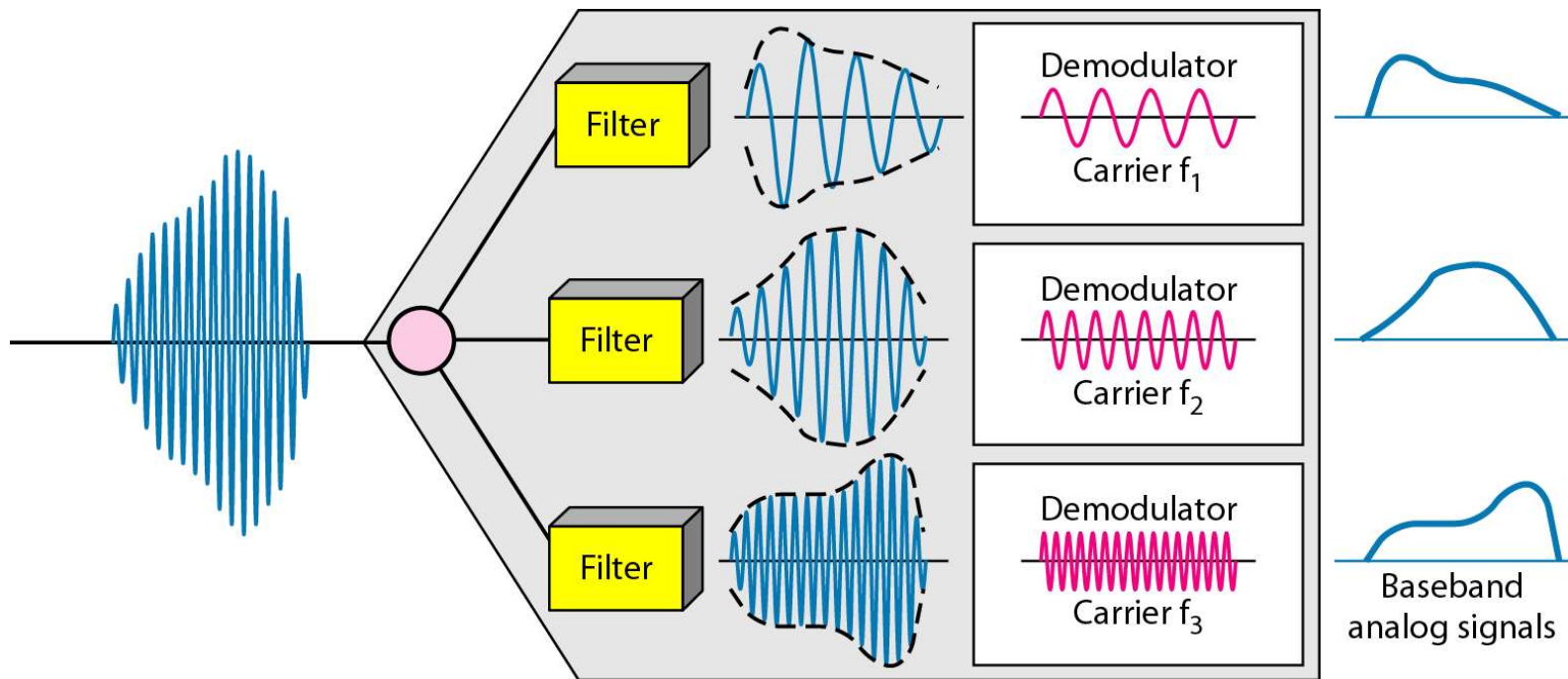
FDM process



Frequency Modulation

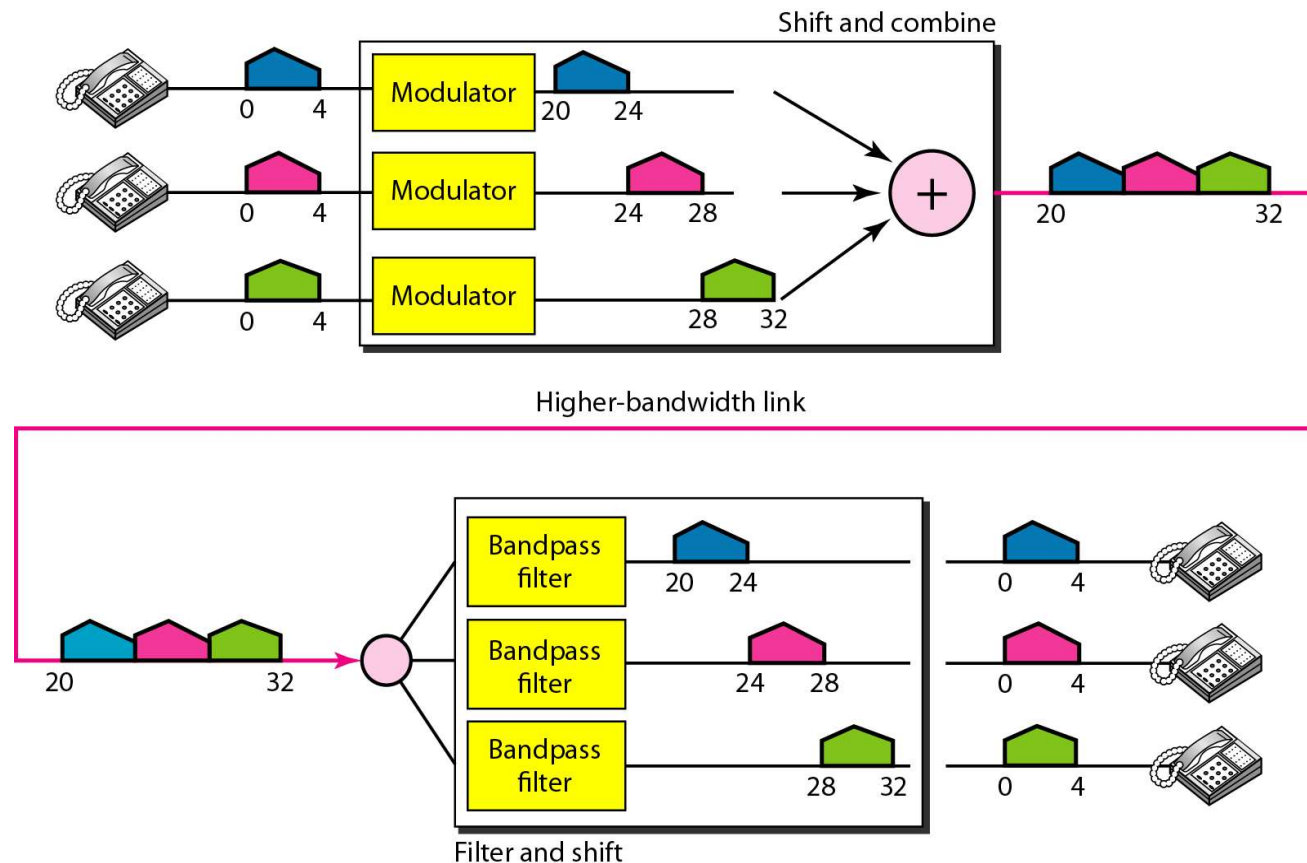


FDM demultiplexing example



FREQUENCY DIVISION MULTIPLEXING (FDM)

Assume that a voice channel occupies a bandwidth of 4 kHz. We need to combine three voice channels into a link with a bandwidth of 12 kHz, from 20 to 32 kHz. Show the configuration, using the frequency domain. Assume there are no guard bands.



Five channels, each with a 100-kHz bandwidth, are to be multiplexed together. What is the minimum bandwidth of the link if there is a need for a guard band of 10 kHz between the channels to prevent interference?

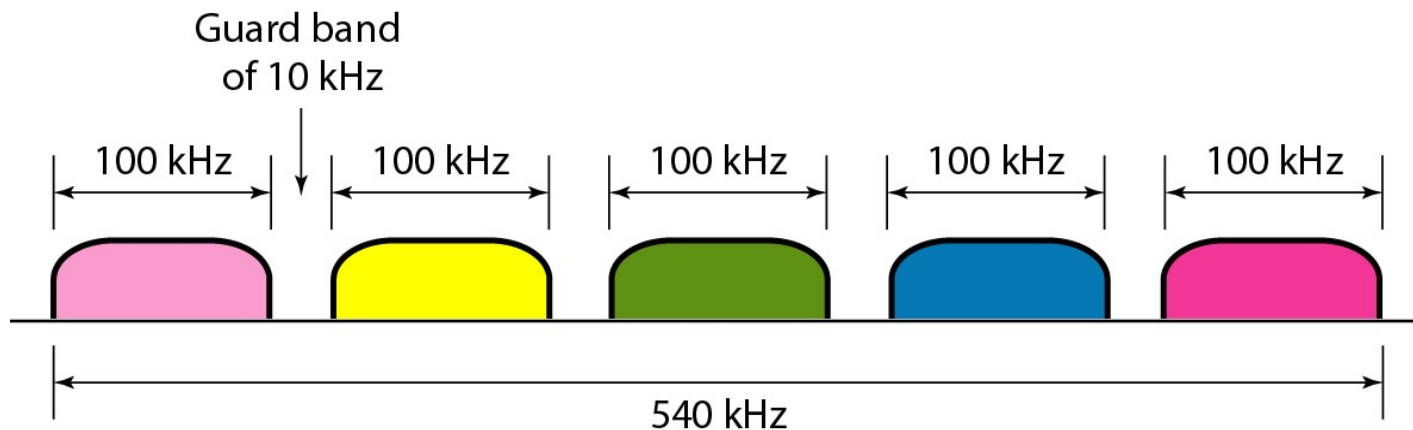
Solution

For five channels, we need at least four guard bands.

This means that the required bandwidth is at least

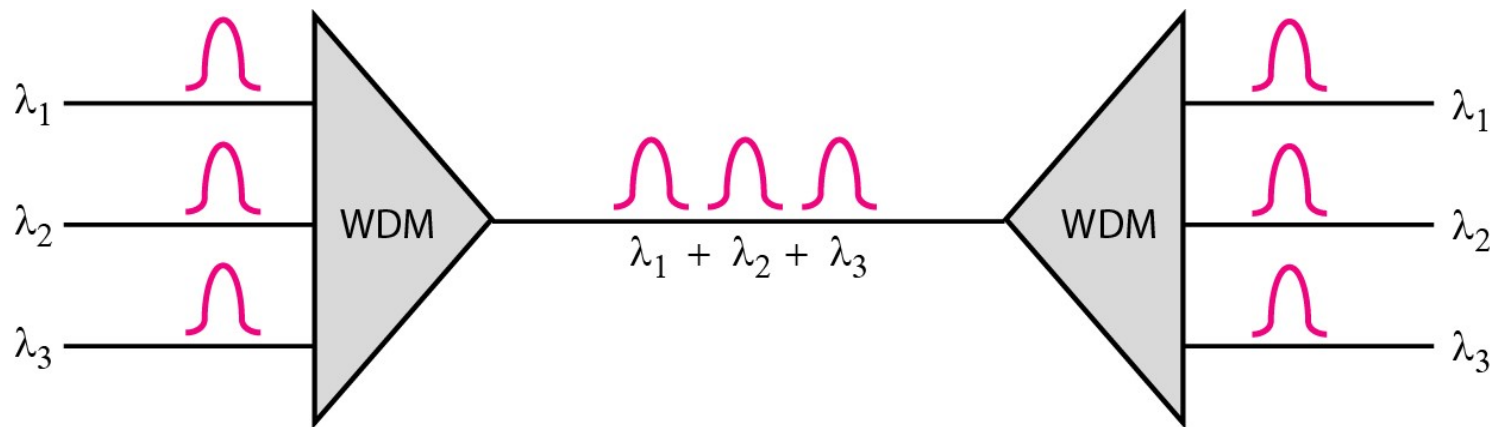
$$5 \times 100 + 4 \times 10 = 540 \text{ kHz},$$

as shown in Figure

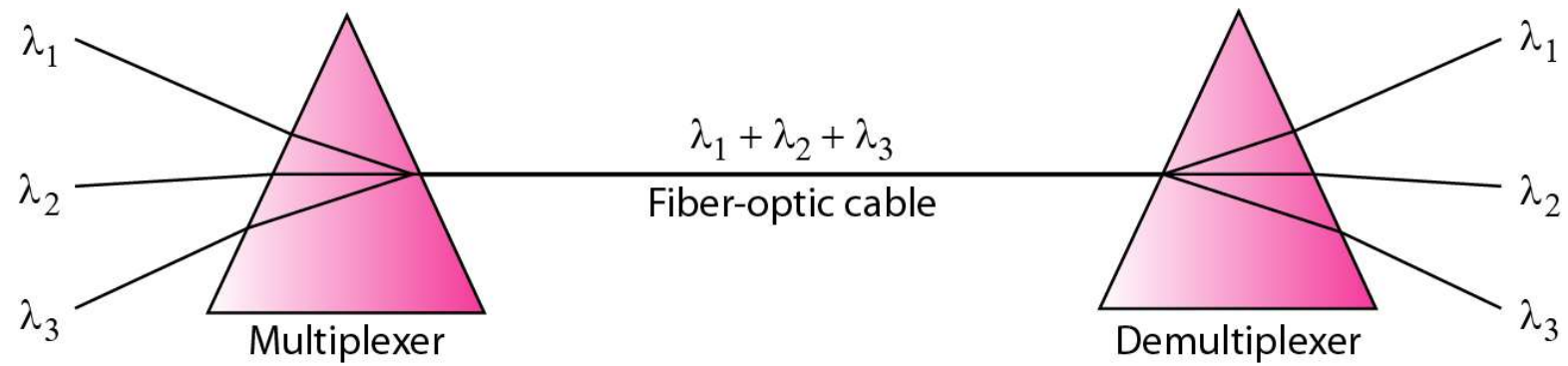


WAVELENGTH DIVISION MULTIPLEXING (WDM)

- This is an analog multiplexing technique to combine optical signals.

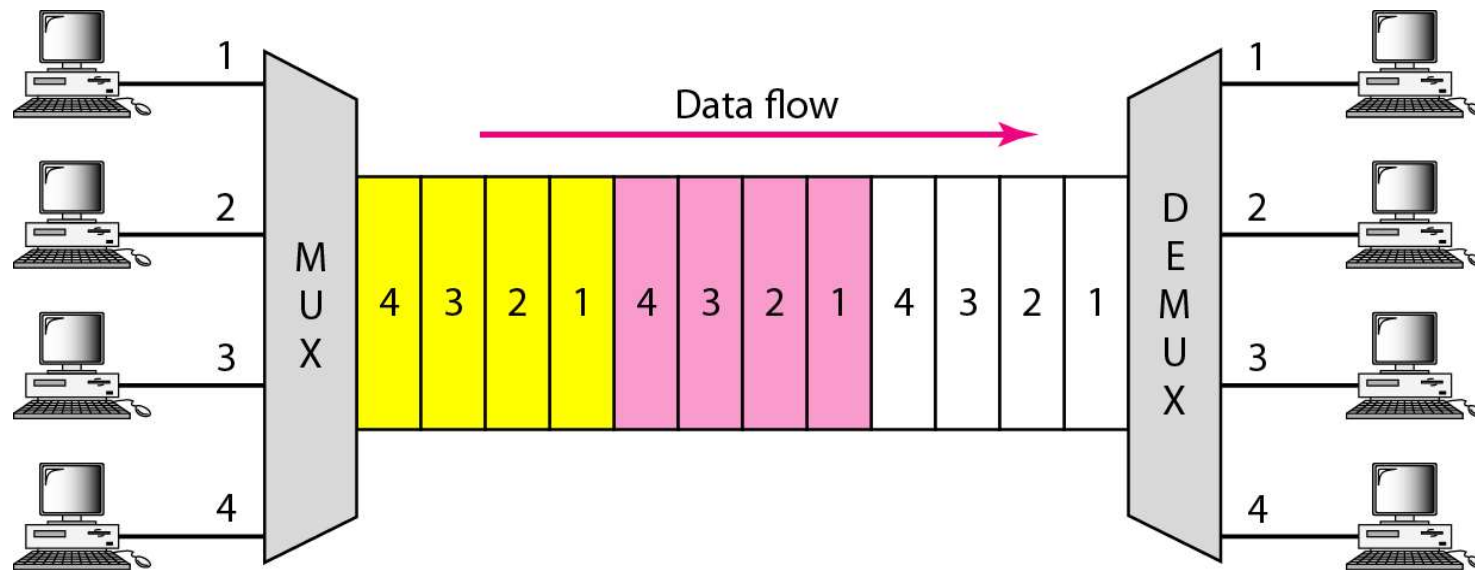


Prisms in wavelength-division multiplexing and demultiplexing



TIME DIVISION MULTIPLEXING (TDM)

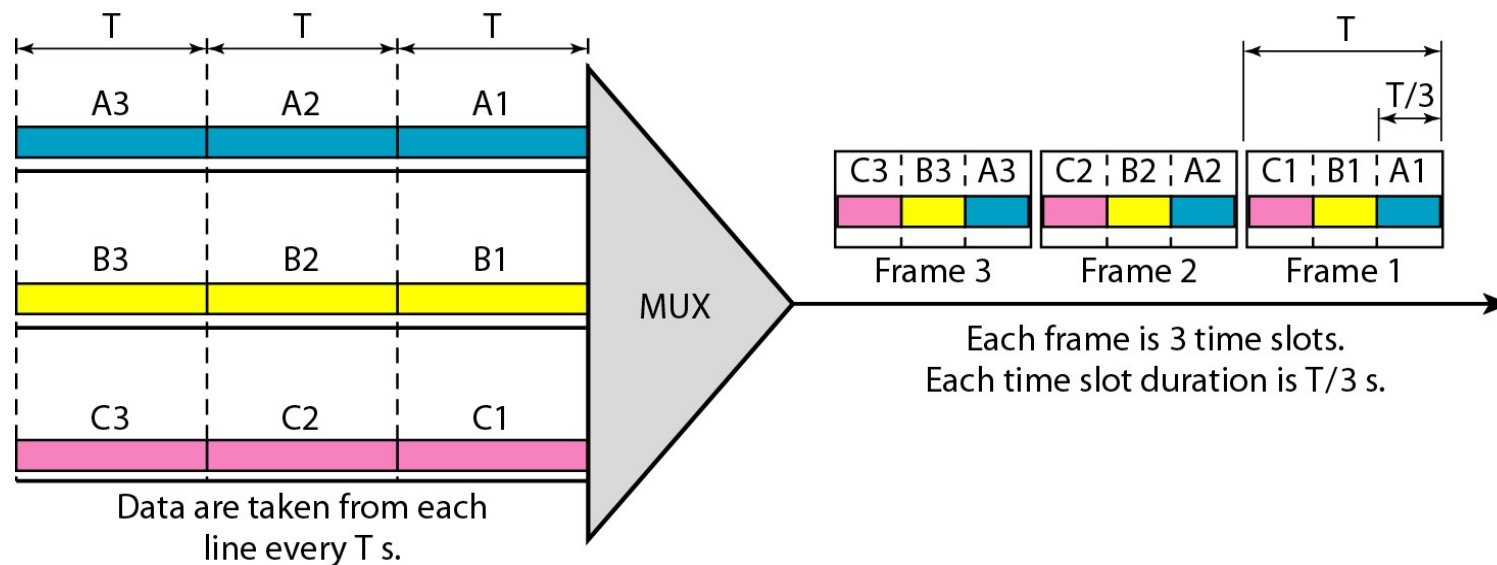
- Instead of sharing a portion of the bandwidth as in FDM, time is shared.
- Analog data can be changed to digital data, and then multiplexed by using TDM.
- TDM is a digital multiplexing technique for combining several low-rate channels into one high-rate one.



TIME DIVISION MULTIPLEXING (TDM)

Time Slots and Frames

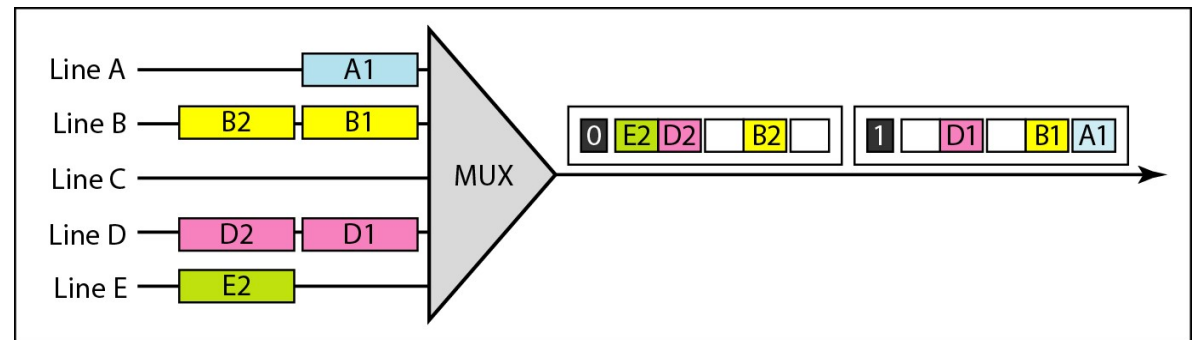
- In Synchronous TDM, the data flow of each input connection is divided into units, where each input occupies one input time slot.
- The duration of an output time slot is n times shorter than the duration of an input time slot.



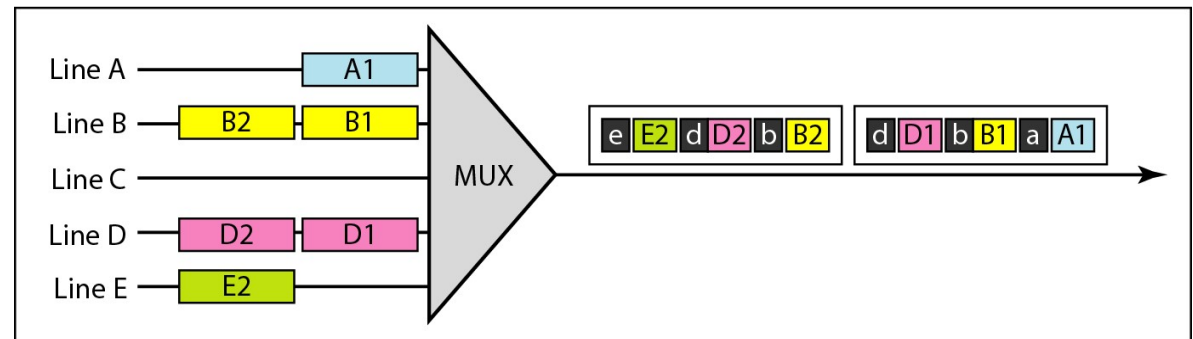
TDM slot comparison

- The **asynchronous time division multiplexing**, which is also called statistical TDM, allocates the time slots dynamically to the data streams based on requirements.

- In such types of multiplexing, the time slot is assigned only when the data stream has some information to be sent, making it more flexible.



a. Synchronous TDM



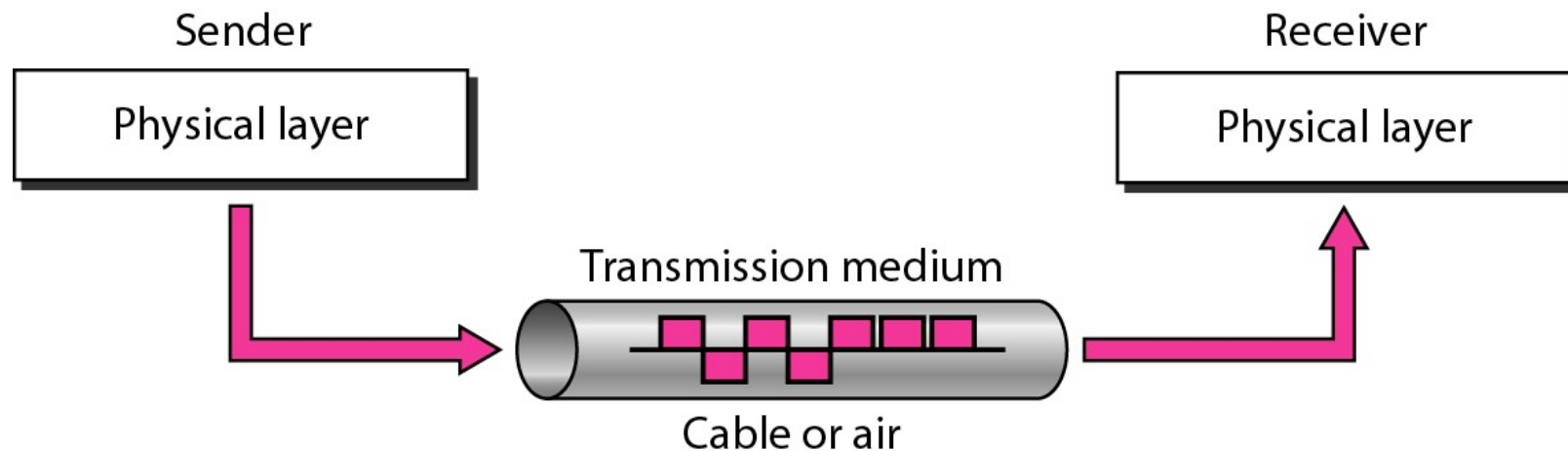
b. Statistical TDM

Feature	Synchronous TDM	Asynchronous TDM (Statistical TDM)
Time Slot Allocation	Fixed time slots pre-assigned to each data stream	Time slots are dynamically assigned based on demand
Bandwidth Utilization	Inefficient; unused slots lead to bandwidth wastage	Efficient; slots are only used when data is present
Data Transmission	Consistent, regardless of whether data is available	Transmits only when data is ready to be sent
Suitability	Ideal for continuous, predictable data streams	Suitable for bursty or irregular data transmission
Latency	Predictable latency due to fixed slots	Latency may vary based on demand and slot availability
Complexity	Simpler to implement with pre-determined slots	More complex due to dynamic slot assignment
Synchronization	Easier to synchronize due to fixed time slots	Harder to synchronize with dynamic allocation
Cost Efficiency	Less cost-efficient due to potential unused slots	More cost-efficient, as resources are allocated on demand

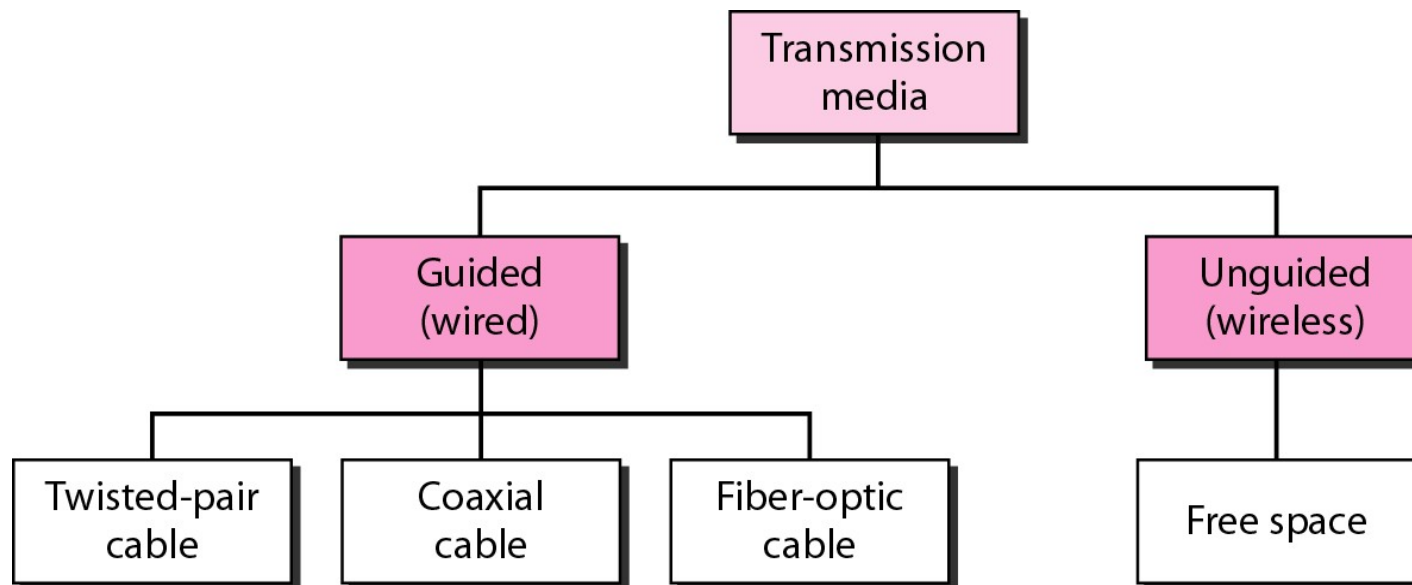
TRANSMISSION MEDIA

Transmission Medium

- A transmission medium can be broadly defined as anything that can carry information from a source to a destination.
- The transmission medium is usually free space, metallic cable, or fiber-optic cable.



Types of Transmission Medium



Guided Media

Guided media, which are those that provide a conduit from one device to another, include twisted-pair cable, coaxial cable, and fiber-optic cable.

Types of Guided Media

Twisted-Pair Cable

Coaxial Cable

Fiber-Optic Cable

Coaxial Cable



Twisted Pair



Fiber Optic

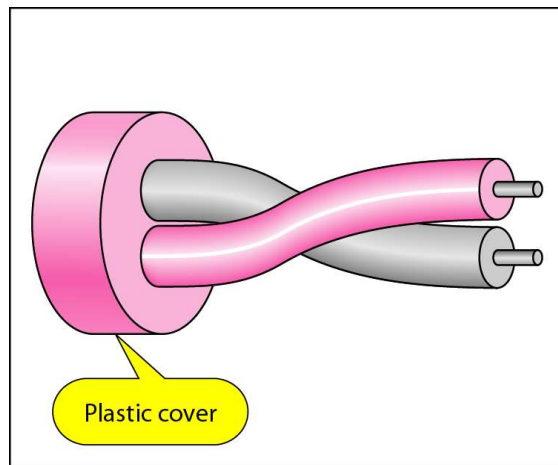


Twisted-Pair Cable

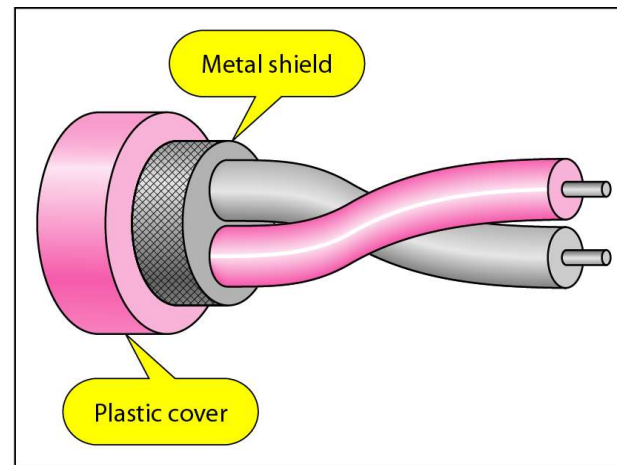
Twisted Pair Cable : Wires are twisted together in pairs. Each pair would consist of a wire used for the positive data signal and a wire used for the negative data signal. Consists of pairs of insulated copper wires twisted around each other.

Unshielded Twisted Pair (UTP): Cables without shields

Shielded Twisted Pair (STP): Cables with shields



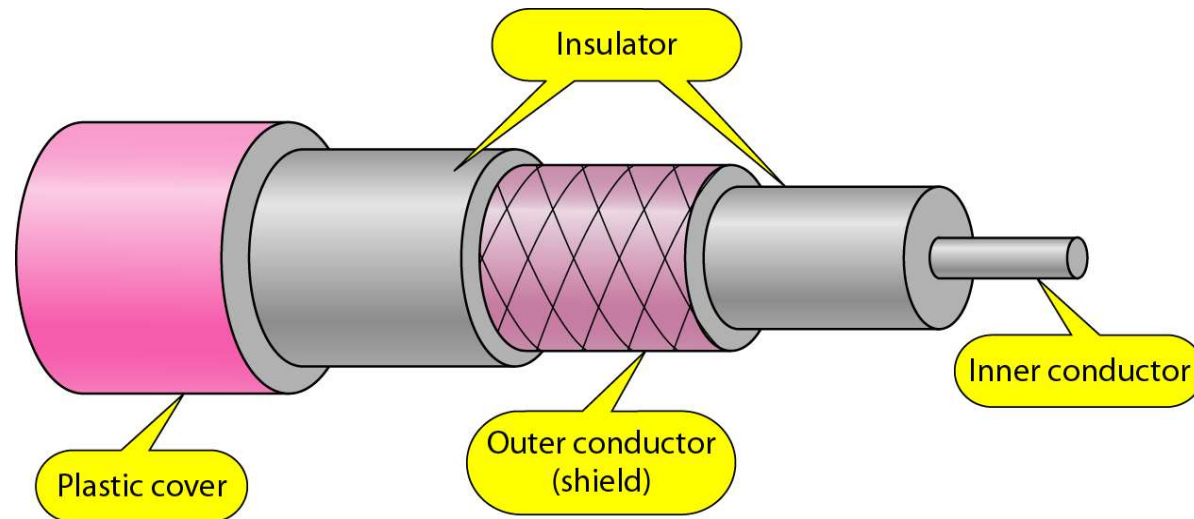
a. UTP



b. STP

Coaxial Cable

- It consists of two conductors. The inner conductor of the [coaxial cable](#) is contained inside the insulator with the other conductor weaves around it providing a shield.
- Consists of copper core conductor, dielectric insulator, metallic shield, and outer cover.
- Provides **better shielding** than twisted pair, hence higher bandwidth.
- **Applications:** Cable TV, broadband Internet, early computer networks.



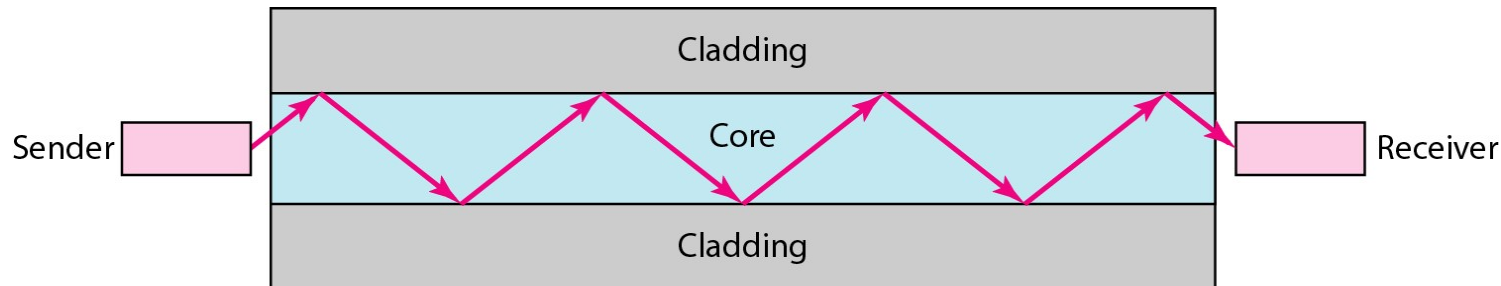
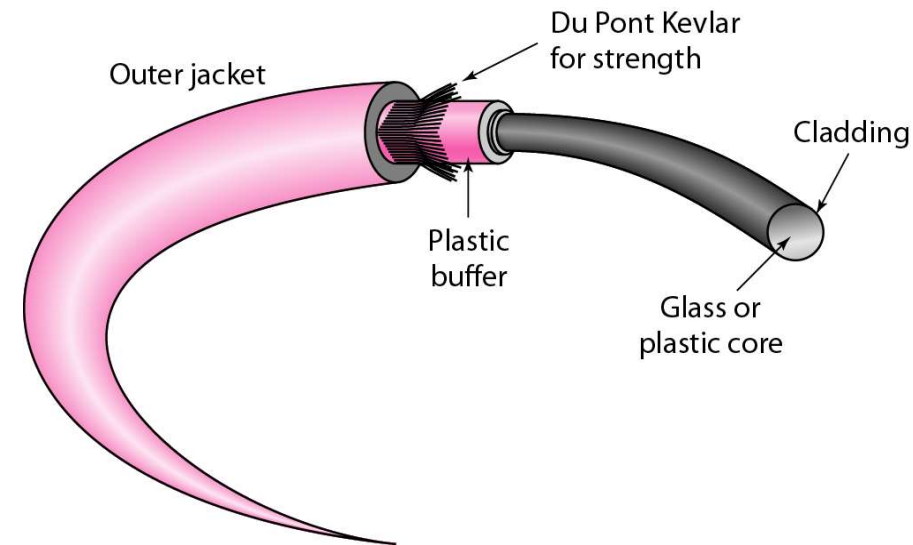
Fiber-Optic Cable

Made of thin strands of **glass or plastic** that transmit data using **light signals**.

Uses **total internal reflection** to confine light inside the core.

Advantages:

- Very high bandwidth (Tbps range).
- Immune to electromagnetic interference.
- Long-distance communication with low attenuation.



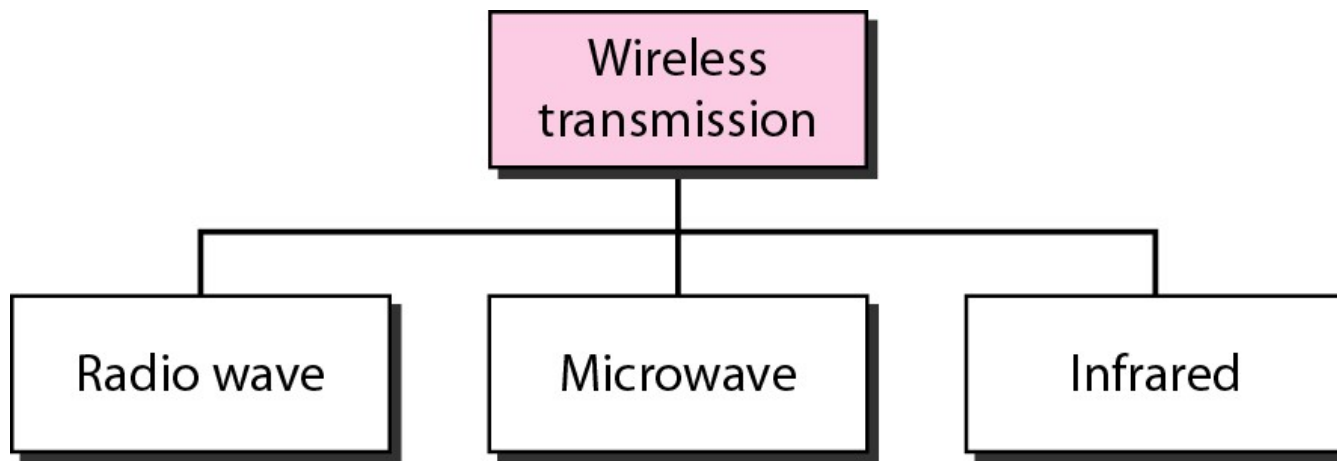
total internal reflection

Guided Media Comparison Table

Medium	Attenuation	Electromagnetic Interface	Security	Cost
Unshielded Twisted Pair	High	High	Low	Low
Shielded Twisted Pair	High	Moderate	Low	Moderate
Coaxial Cable	Moderate	Moderate	Low	Moderate
Fibre Optic Cable	Low	Low	High	High

Unguided Media

Unguided media transport electromagnetic waves without using a physical conductor. This type of communication is often referred to as wireless communication.



Wireless Communication

Radio Waves: Radio waves are used for multicast communications, such as radio and television, and paging systems.

They can penetrate through walls. Use omni directional antennas.

Microwaves: Microwaves are used for unicast communication such as cellular telephones, satellite networks, and wireless LANs.

Higher frequency ranges cannot penetrate walls. Use directional antennas - point to point line of sight communications.

Infrared: Infrared signals can be used for short-range communication in a closed area using line-of-sight propagation.

